INSTITUT UNIVERSITAIRE DES SCIENCES (IUS)

Faculté des sciences et des technologies (FST)

Rapport laboratoire TD N^{O} 1 – Réseaux 2

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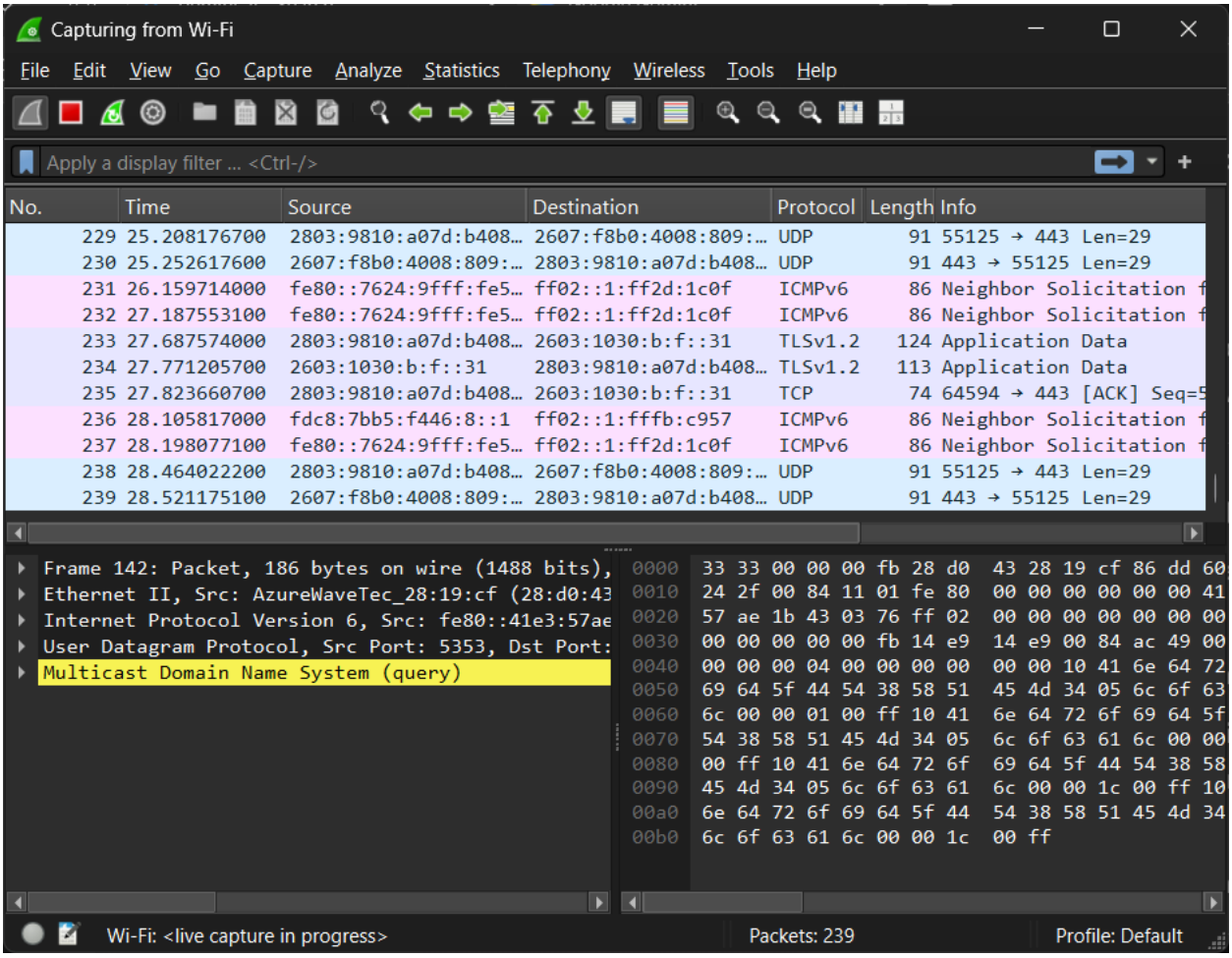
Présenté à: M. Ismaël SAINT AMOUR

Niveau : L3

Date: 22/03/2026

❖ On a l'interface du Wireshark

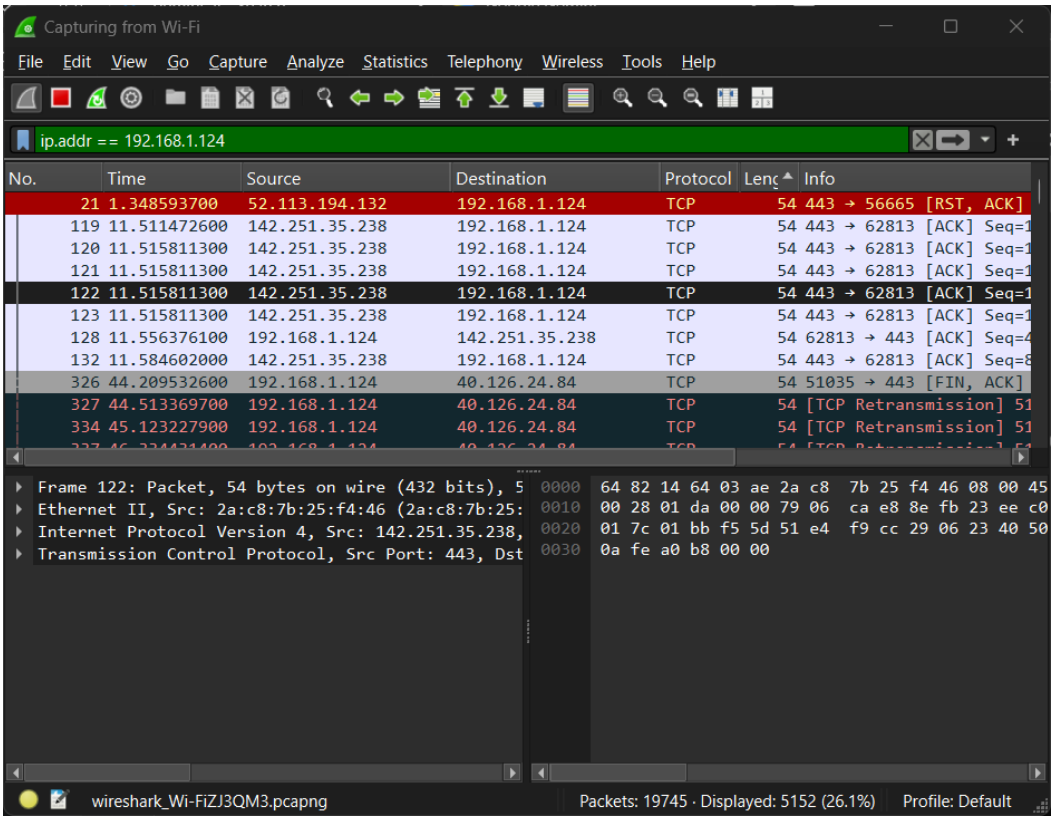


❖ IP Adress dans Powershell : ipconfig

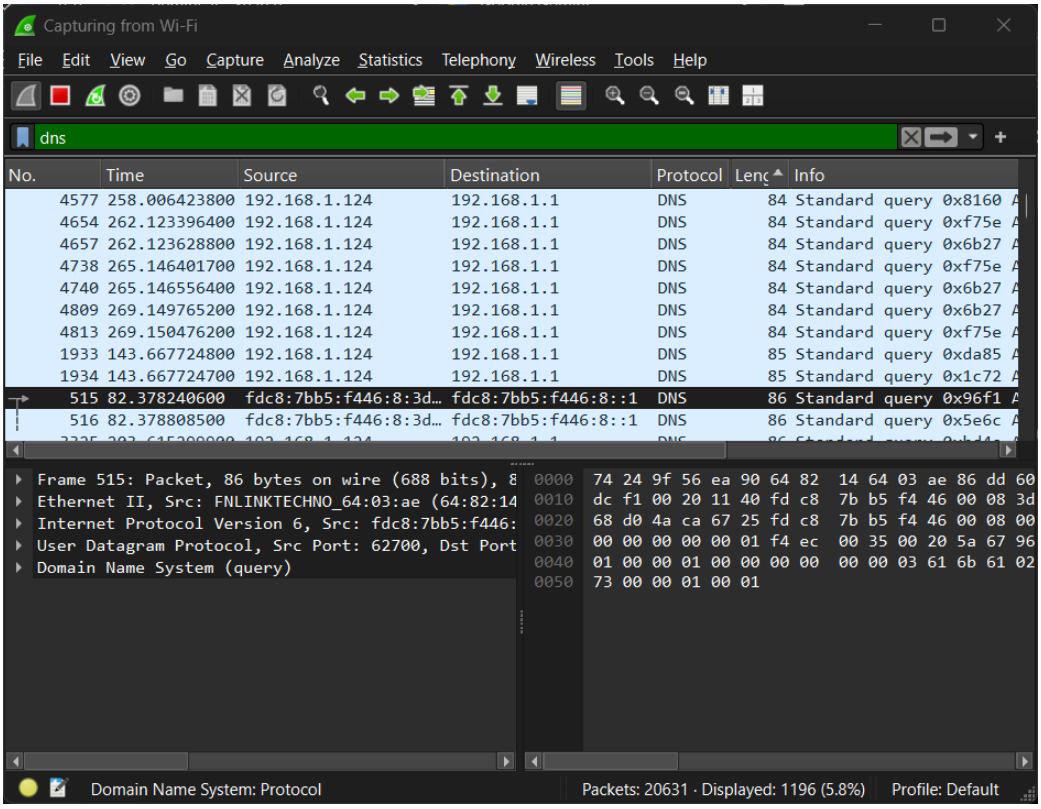
```
Windows PowerShell

PS C:\Users\Brian ANTOINE> ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::6b7a:2a17:553d:8973%20
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : lan
   IPv6 Address. . . . . . . . . . . : 2803:9810:a07d:b408:bfbf:3be5:3098:9855
   IPv6 Address. . . . . . . . . . . : fdc8:7bb5:f446:8:4bba:a699:f4a9:af55
   Temporary IPv6 Address. . . . . . : 2803:9810:a07d:b408:3da5:68d0:4aca:6725
   Temporary IPv6 Address. . . . . . : fdc8:7bb5:f446:8:3da5:68d0:4aca:6725
```

```
Windows PowerShell

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::6b7a:2a17:553d:8973%20
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : lan
   IPv6 Address. . . . . . . . . . . : 2803:9810:a07d:b408:bfbf:3be5:3098:9855
   IPv6 Address. . . . . . . . . . . : fdc8:7bb5:f446:8:4bba:a699:f4a9:af55
   Temporary IPv6 Address. . . . . . : 2803:9810:a07d:b408:3da5:68d0:4aca:6725
   Temporary IPv6 Address. . . . . . : fdc8:7bb5:f446:8:3da5:68d0:4aca:6725
   Link-local IPv6 Address . . . . . : fe80::cf4d:9025:a2f2:29f0%7
   IPv4 Address. . . . . . . . . . . : 192.168.1.124
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::7624:9fff:fe56:ea90%7
                                       192.168.1.1
PS C:\Users\Brian ANTOINE>
```

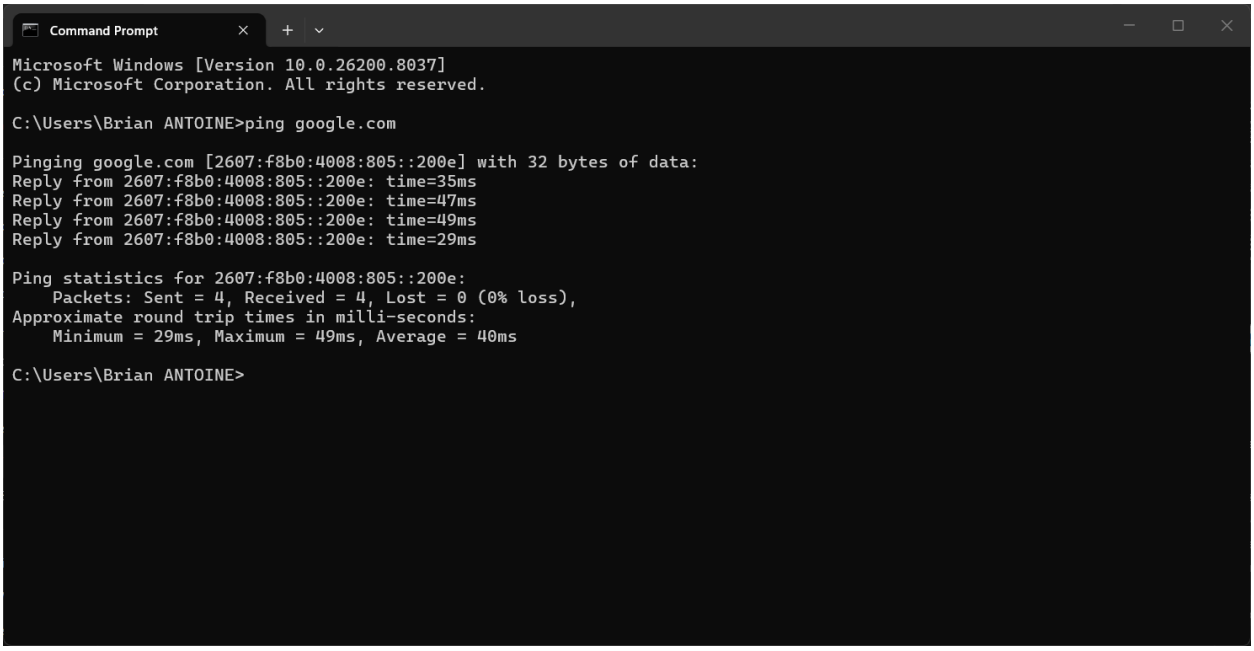
❖ Avec mon adresse ip sur wireshark :


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.addr == 192.168.1.124
No. Time Source Destination Protocol Leng Info
21 1.348593700 52.113.194.132 192.168.1.124 TCP 54 443 → 56665 [RST, ACK]
119 11.511472600 142.251.35.238 192.168.1.124 TCP 54 443 → 62813 [ACK] Seq=1
120 11.515811300 142.251.35.238 192.168.1.124 TCP 54 443 → 62813 [ACK] Seq=1
121 11.515811300 142.251.35.238 192.168.1.124 TCP 54 443 → 62813 [ACK] Seq=1
122 11.515811300 142.251.35.238 192.168.1.124 TCP 54 443 → 62813 [ACK] Seq=1
123 11.515811300 142.251.35.238 192.168.1.124 TCP 54 443 → 62813 [ACK] Seq=1
128 11.556376100 192.168.1.124 142.251.35.238 TCP 54 62813 → 443 [ACK] Seq=4
132 11.584602000 142.251.35.238 192.168.1.124 TCP 54 443 → 62813 [ACK] Seq=8
326 44.209532600 192.168.1.124 40.126.24.84 TCP 54 51035 → 443 [FIN, ACK]
327 44.513369700 192.168.1.124 40.126.24.84 TCP 54 [TCP Retransmission] 51
334 45.123227900 192.168.1.124 40.126.24.84 TCP 54 [TCP Retransmission] 51
Frame 122: Packet, 54 bytes on wire (432 bits), 5
Ethernet II, Src: 2a:c8:7b:25:f4:46 (2a:c8:7b:25:
Internet Protocol Version 4, Src: 142.251.35.238,
Transmission Control Protocol, Src Port: 443, Dst
0000 64 82 14 64 03 ae 2a c8 7b 25 f4 46 08 00 45
0010 00 28 01 da 00 00 79 06 ca e8 8e fb 23 ee c0
0020 01 7c 01 bb f5 5d 51 e4 f9 cc 29 06 23 40 50
0030 0a fe a0 b8 00 00
wireshark_Wi-FiZJ3QM3.pcapng
Packets: 19745 · Displayed: 5152 (26.1%)
Profile: Default

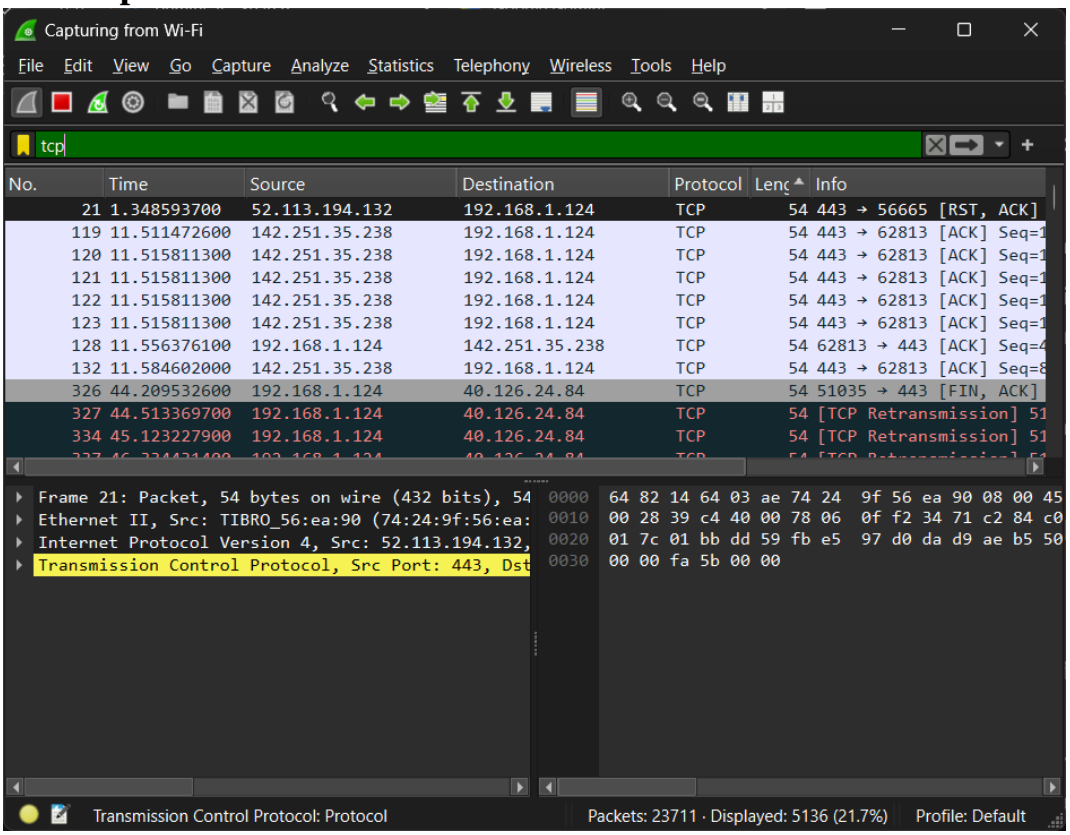
❖ Pour dns :


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Leng Info
4577 258.006423800 192.168.1.124 192.168.1.1 DNS 84 Standard query 0x8160
4654 262.123396400 192.168.1.124 192.168.1.1 DNS 84 Standard query 0xf75e
4657 262.123628800 192.168.1.124 192.168.1.1 DNS 84 Standard query 0x6b27
4738 265.146401700 192.168.1.124 192.168.1.1 DNS 84 Standard query 0xf75e
4740 265.146556400 192.168.1.124 192.168.1.1 DNS 84 Standard query 0x6b27
4809 269.149765200 192.168.1.124 192.168.1.1 DNS 84 Standard query 0x6b27
4813 269.150476200 192.168.1.124 192.168.1.1 DNS 84 Standard query 0xf75e
1933 143.667724800 192.168.1.124 192.168.1.1 DNS 85 Standard query 0xda85
1934 143.667724700 192.168.1.124 192.168.1.1 DNS 85 Standard query 0x1c72
515 82.378240600 fdc8:7bb5:f446:8:3d… fdc8:7bb5:f446:8::1 DNS 86 Standard query 0x96f1
516 82.378808500 fdc8:7bb5:f446:8:3d… fdc8:7bb5:f446:8::1 DNS 86 Standard query 0x5e6c
Frame 515: Packet, 86 bytes on wire (688 bits), 8
Ethernet II, Src: FNLINKTECHNO_64:03:ae (64:82:14
Internet Protocol Version 6, Src: fdc8:7bb5:f446:
User Datagram Protocol, Src Port: 62700, Dst Port
Domain Name System (query)
0000 74 24 9f 56 ea 90 64 82 14 64 03 ae 86 dd 60
0010 dc f1 00 20 11 40 fd c8 7b b5 f4 46 00 08 3d
0020 68 d0 4a ca 67 25 fd c8 7b b5 f4 46 00 08 00
0030 00 00 00 00 00 01 f4 ec 00 35 00 20 5a 67 96
0040 01 00 00 01 00 00 00 00 00 00 03 61 6b 61 02
0050 73 00 00 01 00 01
Domain Name System: Protocol
Packets: 20631 · Displayed: 1196 (5.8%)
Profile: Default

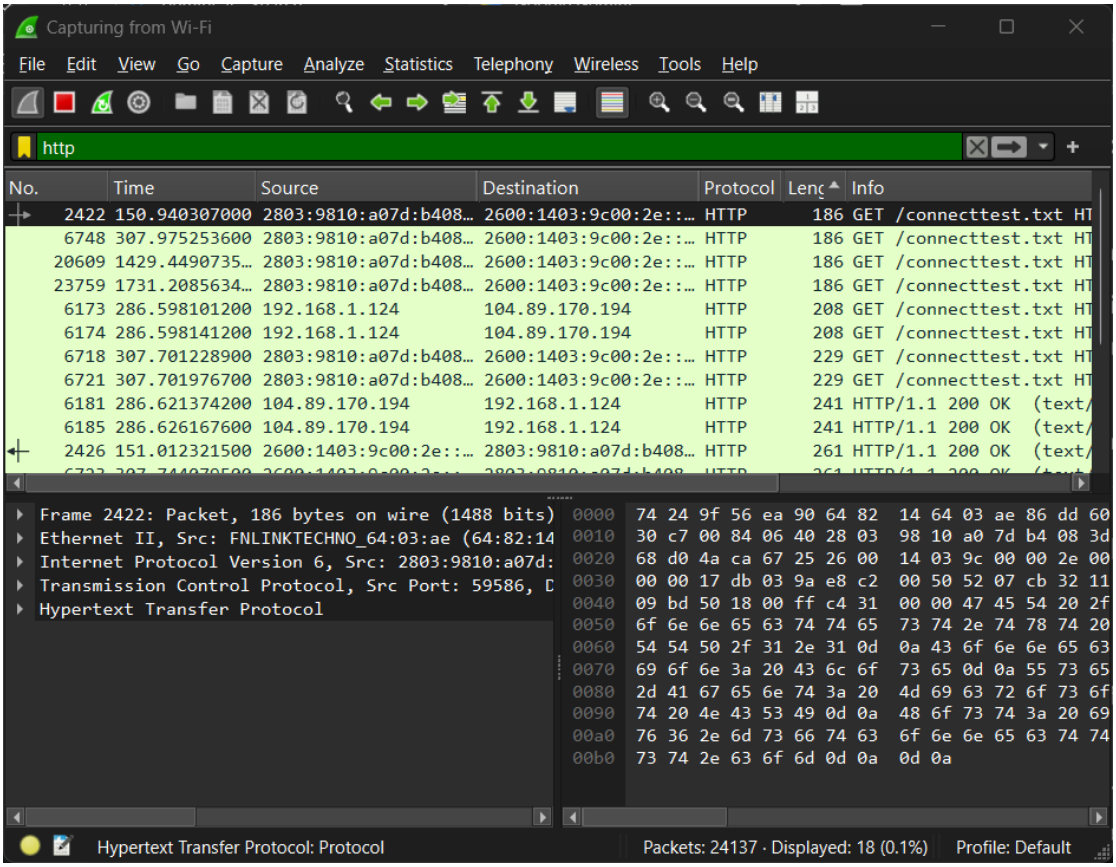
❖ Faisons ping google.com dans CMD



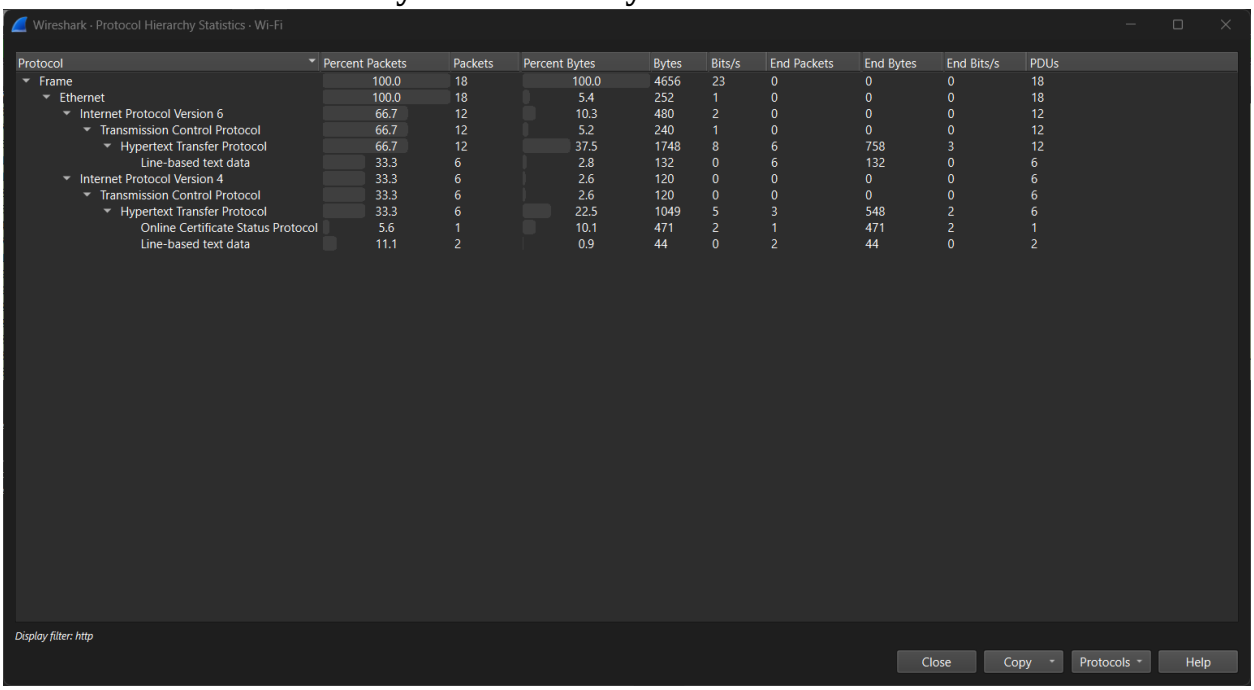
❖ Pour tcp



Pour http :

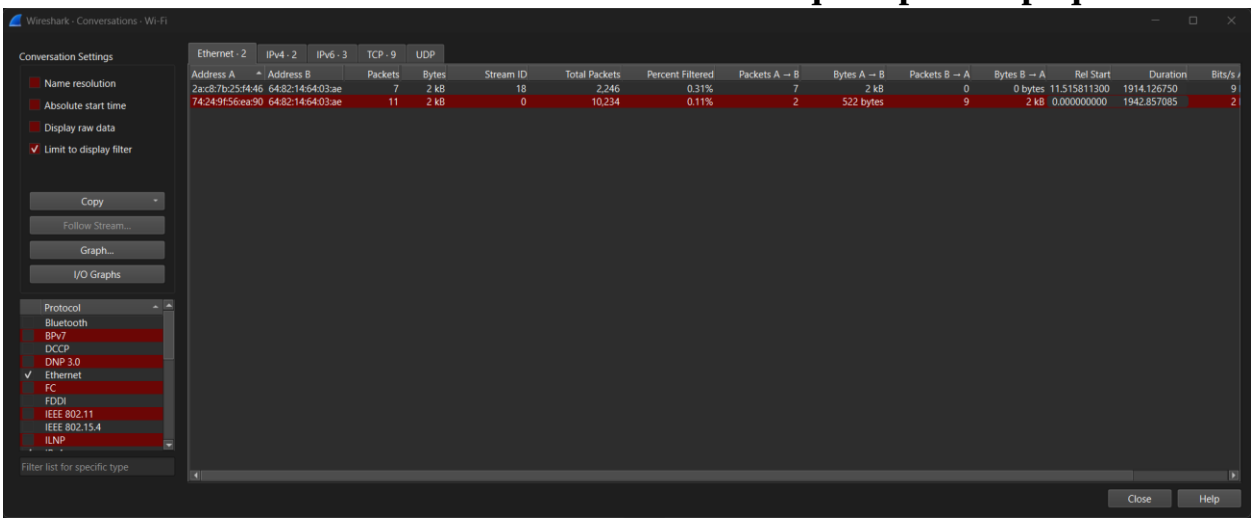


❖ ON ouvre Statistics >Ptoyocol hierarchy

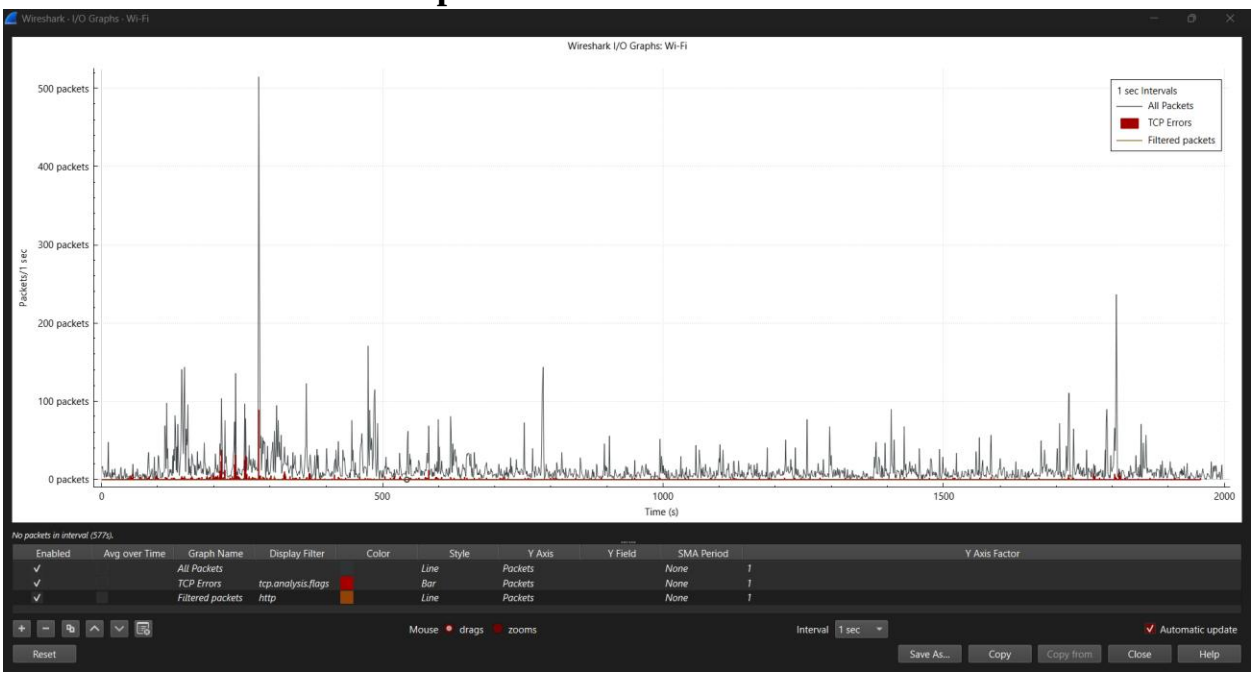


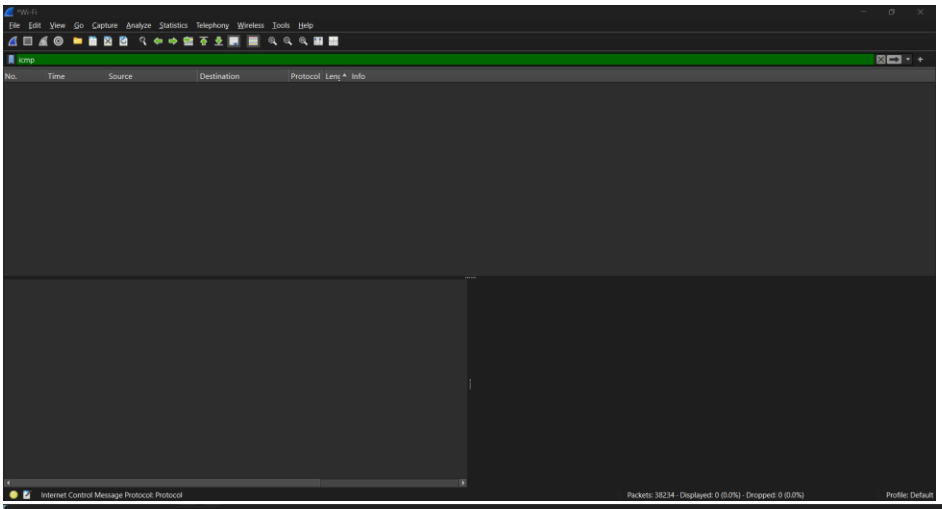
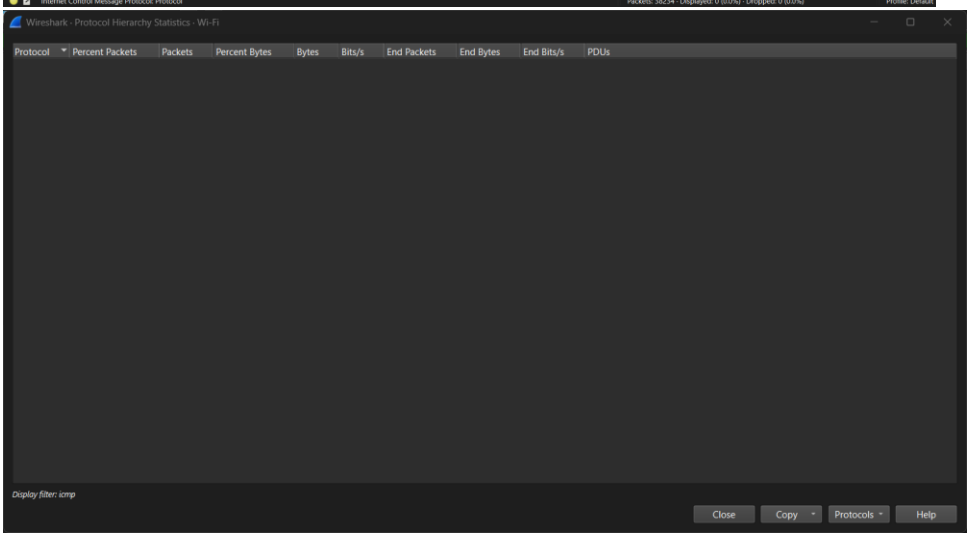
Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	18	100.0	4656	23	0	0	0	18
Ethernet	100.0	18	5.4	252	1	0	0	0	18
Internet Protocol Version 6	66.7	12	10.3	480	2	0	0	0	12
Transmission Control Protocol	66.7	12	5.2	240	1	0	0	0	12
Hypertext Transfer Protocol	66.7	12	37.5	1748	8	6	758	3	12
Line-based text data	33.3	6	2.8	132	0	6	132	0	6
Internet Protocol Version 4	33.3	6	2.6	120	0	0	0	0	6
Transmission Control Protocol	33.3	6	2.6	120	0	0	0	0	6
Hypertext Transfer Protocol	33.3	6	22.5	1049	5	3	548	2	6
Online Certificate Status Protocol	5.6	1	10.1	471	2	1	471	2	1
Line-based text data	11.1	2	0.9	44	0	2	44	0	2

❖ **On ouvre Statistics> conversations>identifier l'IP qui le plus de paquets**



❖ **Ouvrons statistics>I/O Graphs >>**



Inconvénients :

L'objectif de ce TD était de comprendre le rôle de Wireshark dans la capture et l'analyse du trafic réseau. Lors de l'exécution, j'ai pu identifier avec succès les sessions TCP (handshake SYN/ACK) et les requêtes HTTP. Cependant, concernant le protocole ICMP (utilisé pour le ping), aucun paquet n'a été détecté dans la capture malgré l'envoi des commandes. Cette absence s'explique par un filtrage rigoureux du pare-feu de la machine cible ou du réseau, qui bloque les requêtes Echo Request pour prévenir les scans de ports et les intrusions. Ce constat illustre concrètement l'un des points clés du TD : comprendre comment les configurations réseau permettent de détecter ou d'empêcher des comportements suspects. Enfin, l'utilisation des outils statistiques comme le Protocol Hierarchy et les I/O Graphs reste essentielle pour visualiser tout pic de trafic anormal, confirmant ainsi la réussite de l'apprentissage des méthodes d'analyse de sécurité.